

READING NOTES

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1. INTRODUCTION

The motivation behind this running record of reading notes on mathematics is to gather the ongoing observations, ideas and questions which arise during my study. The entries are primarily for my own understanding, but I hope they may also be of interest to others who enjoy exploring mathematics. As my understanding evolves, some notes may be revised in the future.

2. NOTES

05th Aug, 2026. One of the more surprising ideas I encountered today was that the common notion of the zero set of a collection of polynomials is elevated to a central geometric object in algebraic geometry. Such a set is called an *algebraic set*, while a *variety* is, roughly speaking, an irreducible algebraic set (depending on the convention being used).

I also found it interesting that these objects are studied in affine space $(\mathbb{A}_k^n)$. Although affine space has points with coordinates such as $(0, 0)$ or, more generally, $(0, \dots, 0)$, it is not regarded as having a distinguished origin in the geometric sense. The point $(0, \dots, 0)$ is then simply another point of the space, and whether it belongs to the zero set of a polynomial depends entirely on the polynomial itself. For instance, if the constant term is nonzero, then $(0, \dots, 0)$ need not be a zero of that polynomial.

Another pleasing observation was that the Cayley graph of the free group F_2 , with respect to its standard free generators, is the infinite 4-regular tree T_4 . Starting from any vertex, the tree branches outward in a remarkably symmetric fashion, and after the initial four edges, there are $4 \cdot 3^{n-1}$ new vertices at distance n from the starting point.

Moreover, algebraic sets are interesting objects. Although they can be defined over any field, many of their fundamental properties are studied over algebraically closed fields. There is also the *Zariski topology*, which arises naturally on affine spaces and whose closed sets are precisely the algebraic sets, at least for $n = 1$ and $n = 2$.

04th Aug, 2026. Well, the Cayley graph of a free group with respect to a free basis is an infinite regular tree. This graph has some interesting valencies at each point and it is unlike the finite groups' graphs which contain cycles. In other news, $SL_2(\mathbb{Z})$ has some interesting elements which are also its generators. It would be interesting to explore the geometry of its Cayley graph.

03rd Aug, 2026. Today I learned that automorphisms of a group induce automorphisms of its Cayley graph (provided the generating set is preserved). This is interesting because every automorphism of a group is a permutation of its underlying set, so there is a natural embedding $\text{Aut}(G) \hookrightarrow \text{Sym}(G)$. Thus, the same automorphism acts both as a permutation of the group and as a symmetry of its Cayley graph.